Field-based evaluation of sugarcane bagasse biochar for managing root-knot nematodes in tomato

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Highlights

- Root-knot nematodes cause major yield losses in tomato crops worldwide.

- Sugarcane biochar at 30g/plant reduced RKN infection by 76 % vs. control.

- SC-biochar upregulated JERF-3 and PR-1 defense genes in tomato roots.

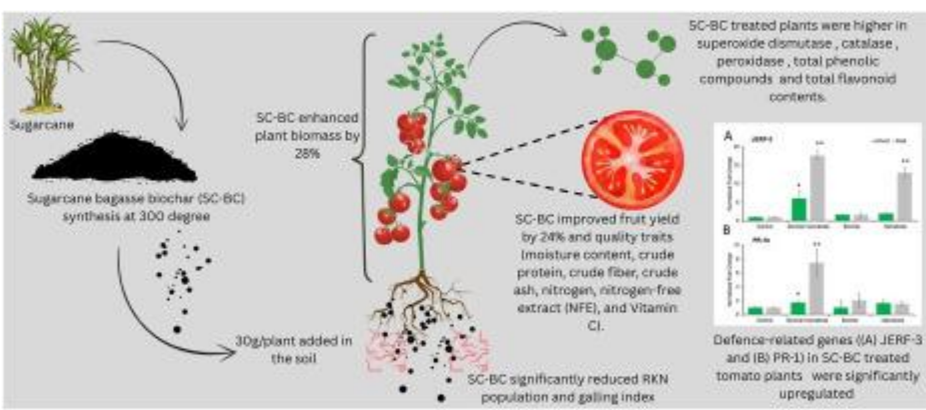
- Antioxidants and phenolic compounds increased in biochar-treated plants.

Abstract

Root-knot nematodes (RKN) from *Meloidogyne* genus cause significant yield losses in tomato crops globally. Chemical nematicides used to control them could be detrimental to the human beings and the environment. Therefore, the present study was designed to manage RKN infestations in tomatoes by using biochar developed from sugarcane bagasse. Sugarcane bagasse-derived biochar (SC-BC) pyrolyzed at 300 °C amended in the soil at 30g per plant significantly reduced RKN infection compared to the control. Plants treated with biochar and nematodes (BC + N+) showed a 76.3 % reduction in galling index compared to nematode-only plants. Root galling was significantly higher in nematode-only plants (mean GI = 4.7) than in BC + N+ (mean GI = 1.1). The current findings indicate that the 30 g per plant sugarcane bagasse biochar, tested in the field showed promising results. This treatment increased plant height, a higher quantity of fruit, and a reduction in

nematode populations compared to the control group. This study analyzed fruit quality traits, including moisture content, crude protein, crude fiber, crude ash, nitrogen, nitrogen-free extract (NFE), and Vitamin C. While moisture content, crude ash, nitrogen, NFE, and crude protein remained unchanged in all treatments. Biochar-treated plants exhibited 28 % higher shoot biomass, and 24 % more fruit yield compared to untreated nematode-infected plants. The chemical control produced similar yields but lacked improvements in plant vigor. Crude fiber and vitamin C levels were highest in the nematode-only group, likely as a stress-induced metabolic response. BC+N+ treatment restored nutrient balance and improved fruit firmness and sugar content. The study revealed that sugarcane bagasse in nematode-infested treatment (BC-N+) activated defence-related genes in tomato plants, where both JERF-3 and PR-1 genes were upregulated significantly (17.65 and 7.46 folds, respectively) in roots of tomato compared to control. Biochemical enzymes such as superoxide dismutase (SOD), catalase (CAT), peroxidase (POD), total phenolic compounds (TPC) and total flavonoid contents (TFC) were higher in biochar (BC + N- and BC + N+) treated plants in both roots and shoots. Chlorophyll fluorescence parameters (Fv/Fm) indicated improved photosystem II efficiency in BC+N+ plants under nematode stress, approaching levels seen in the non-infected control (BC-N-). The present study found that using 30g per plant sugarcane BC effectively managed RKN, boosted plant biomass, and potentially activated defence mechanisms in tomatoes.

Graphical abstract



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Introduction

After potatoes, tomatoes (*Solanum lycopersicum* L.) are the 2nd most widely grown vegetable globally [1]. Climate change including increased temperatures, altered irrigation water availability, droughts, soil salinization, reduced soil fertility, has significantly altered the microbial communities of the rhizosphere and higher occurrence of the rhizosphere pests like plant-parasitic nematodes [2,3]. The incidence of plant-parasitic nematodes have been enhanced under drought conditions in different crop plants. These parasitic worms are recognized as one of the greatest threats to crops globally [4]. These worms are devastating parasites of several crops and lead to crop losses of worth USD 152 billion worldwide annually [5]. Among different plant-parasitic nematodes, root-knot nematodes (RKN) (*Meloidogyne* spp.) are considered as the most harmful plant-parasitic nematodes in tomato [6,7]. Their infestation usually leads to higher costs of production due to the extensive use of chemical pesticides [8]. Because of their diversity and broad host range, managing root-knot nematodes, particularly *Meloidogyne incognita* is difficult [9]. To combat these pests, several tactics have been investigated including the use of chemical nematicides, the breeding of resistant crops, the use of farming techniques, and the introduction of biological agents [[10], [11], [12]]. However, use of chemical nematicides creates health and environmental issues which necessitates the search for sustainable alternatives [[13], [14], [15]].

The application of organic amendments, such as biochar (BC) is one of the successful strategies to combat biotic and abiotic stresses [16,17]. Biochar is generally made by burning organic materials in a furnace in the absence of oxygen at high temperatures [18]. Numerous studies have been carried out to investigate that how BC affects several aspects of plant health in response to plant-parasitic nematodes [19]. According to Huang et al. [20], rice plants infected with *M. graminicola* showed activation of defence responses because of BC amendment in a pot experiment. Similarly, when BC was added to wheat field, Zhang et al. [21] found that the number of plant-parasitic nematodes decreased and the population of fungivore nematodes significantly increased. After applying wood BC at a rate of 5 %, another study showed that the infestation of root-lesion nematodes (*Pratylenchus penetrans*) in carrot fields was greatly decreased [22]. Similarly, 5 % (w/w) sugarcane bagasse BC synthesized at 450 °C decreased overall RKN infestation and greatly improved overall plant biomass and physiological functions in tomato plants according to a research by Arshad et al. [7].

Biochar amendments make plants more resistant to a variety of insect pests and pathogens, such as bacteria, fungi, viruses, and nematodes [17,23]. Furthermore, a recent study demonstrated that an aqueous solution of grape pomace BC prepared at 350 °C was effective in controlling *M. javanica* due to direct nematicidal activity and enhanced availability of macro and micronutrients that improved soil beneficial microbial communities [24]. By keeping these studies in mind, the current study was planned to use sugarcane bagasse biochar developed at 300 °C from our pot experiment [25] in the field to devise an eco-friendly method to control RKN in tomato. Additionally, we have studied biochemical parameters including superoxide dismutase (SOD), catalase (CAT), peroxidase (POD), total phenolic compounds (TPC) and total flavonoid contents (TFC), and fruit quality parameters like moisture content, crude ash, nitrogen, NFE, crude fibre and crude protein in response to different biochar and nematode treatments in tomato. Moreover, we studied the expression of two defence-related genes using qPCR to test the activation of plant defence in response to application of sugarcane bagasse biochar.

Assessment of biochar for controlling nematodes under natural field conditions

The effectiveness of biochar in promoting plant development and controlling RKN was investigated in an experiment. Twenty-five days old tomato nursery var., Rio Grande, was transplanted into the nematode sick plots with the following treatments; 30g grinded biochar only (BC+N-) applied per plant, biochar combined with nematodes (BC+N+), plots with only nematodes (BC-N+) and a negative control with no treatment (BC-N-) (Table 1), at the Plant Pathology Department Experimental area at the

Evaluation of nematode resistance among various treatments in the field experiment

The data regarding the galling index was taken from 5 plants in each treatment from each replication. The results revealed that biochar application significantly controlled the nematode infection (where the galling index was 1.4) as compared to the nematode control (galling index 4.7). (Fig. 1).

Influence of biochar application on various plant parameters under field conditions

In the field experiment, where the tomato variety Rio Grande was transplanted with different treatments, parameters like plant height, chlorophyll contents, number of fruits per plant and fruit weight

Discussion

Agricultural ecosystems are seriously threatened by plant diseases which can result in large yield losses. To increase agricultural yields and decrease disease incidences, many organic methods have been developed [42] and there are several advantages to carbon amendment. Plant productivity is eventually raised by the interaction of carbon with soil additions [43]. Biochar (BC) has been applied to soil for fertilization and acclimatisation. Furthermore, pollutants in the rhizosphere can be

Conclusion

The objective of this research was to investigate the impact of biochar on tomato plants' resistance to the RKN. Nematode resistance was assessed using a galling index (0–10) revealing that applying SC-BC-developed at 300 °C significantly reduced nematode infection compared to the nematode control group (BC-N+). In field experiments, plant height, chlorophyll content, number of fruits per plant and fruit weight were recorded. Plants treated with biochar, BC+N+ and the negative control showed

CRediT authorship contribution statement

Maria Fayyaz: Writing – original draft, Investigation. **Amjad Abbas:** Validation, Methodology, Data curation. **Usman Arshad:** Writing – review & editing, Validation, Software, Formal analysis. **Farrukh Azeem:** Writing – review & editing, Visualization, Resources. **Faiz Ahmad Joyia:** Resources, Formal analysis, Data curation. **Abdelaziz Hajjaji:** Writing – review & editing, Validation, Software. **Muhammad Amjad Ali:** Writing – review & editing, Supervision, Resources, Project administration, Funding

Declaration of competing interest

The authors declare that there are no potential conflicts of interest.

Acknowledgements

The authors are thankful to Higher Education Commission of Pakistan to fund this research through National Research Program for Universities grant number NRPU-9087. The authors are highly thankful to Prof. Dr. Muhammad Naveed for providing the facility of Muffle Furnace to pyrolyze the biochar.